

SIMATS ENGINEERING

ASSESSMENT TOOL 1

Experiments

COURSE NAME: Computer Networks

COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO3: *Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions*

Assessment Tool Weightage: 40%

SOFTWARE: CISCO PACKET TRACER, WIRESHARK

EXPERIMENTS

QUESTIONS:4

Total Marks:40

1. Design a simple LAN consisting of two PCs connected through a switch using Cisco Packet Tracer. Configure appropriate IP addresses for both hosts and verify connectivity using the ping command. Simultaneously capture the packet exchange using Wireshark. Identify the Ethernet frame fields such as Source MAC Address, Destination MAC Address, EtherType, and Frame Check Sequence (FCS). Explain how the Data Link layer uses MAC addressing to ensure successful frame delivery within the local network.
2. Create a network topology consisting of two LANs connected through a router in Cisco Packet Tracer. Configure IPv4 addressing, subnet masks, and default gateways for all devices, and verify end-to-end communication using the ping command. Capture the ICMP packets using Wireshark and analyze the IPv4 header fields, including Source IP Address, Destination IP Address, Time-To-Live (TTL), Protocol Number, and Header Length. Explain the role of the Network layer in forwarding packets between different networks.
3. Construct a local area network with multiple hosts connected through a switch in Cisco Packet Tracer. Clear the ARP cache on one of the hosts and initiate communication with another host using the ping command. Capture the communication in Wireshark and identify the ARP Request and ARP Reply packets. Analyze the sender and target MAC/IP addresses and explain how the Address Resolution Protocol resolves logical IP addresses into physical MAC addresses before data transmission.
4. Develop a simple network consisting of two PCs connected through a router in Cisco Packet Tracer. Configure the required IP addresses and test connectivity using the ping command. Capture the ICMP traffic using Wireshark and identify the Echo Request and Echo Reply messages. Examine important ICMP fields such as Type, Code, Identifier, and Sequence Number, and calculate the Round Trip Time (RTT). Discuss how ICMP assists in network troubleshooting and connectivity verification.

5. Create a client-server network in Cisco Packet Tracer by configuring a web server and a client PC. Enable the HTTP service on the server and access the hosted webpage from the client using a web browser. Capture the communication in Wireshark and identify the TCP three-way handshake, followed by the HTTP GET request and HTTP response messages. Analyze the application-layer communication and explain how TCP provides reliable delivery for HTTP traffic through connection establishment, acknowledgments, and ordered data transfer.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

Aim
To design a LAN with two PCs connected through a switch, verify communication using Ping, and analyze Ethernet frames in Wireshark.

- Network Configuration**
- PC1: 192.168.1.10/24
 - PC2: 192.168.1.20/24

- Procedure**
- Connect two PCs to a switch.
 - Configure IP addresses.
 - Execute ping 192.168.1.20.
 - Capture packets in Wireshark.
 - Observe Ethernet frame fields.

Observation

Field	Sample Value
Source MAC	00:1A: 2B:3C:4D:01
Destination MAC	00:1A: 2B:3C:4D:02
EtherType	0x0800 (IPv4)
FCS	Valid

Calculation
Packets Sent = 4
Packets Received = 4
Packet Loss = 0
Packet Loss (%) = (Lost / Sent) × 100
= (0/4) × 100
= **0%**

- Result Analysis**
- Ethernet frame correctly contains Source MAC, Destination MAC and EtherType.
 - Switch forwards frames using MAC addresses.
 - No packet loss indicates successful communication.

Conclusion
The LAN was successfully implemented. MAC addressing enabled correct frame delivery at the Data Link Layer.

Experiment 2: Two LANs Connected by Router (IPv4 Analysis)

Aim
To configure two LANs connected by a router and analyze IPv4 packets.

- IP Configuration**
- LAN1
- PC1: 192.168.1.10/24
 - Gateway: 192.168.1.1

- LAN2
- PC2: 192.168.2.10/24
 - Gateway: 192.168.2.1

- Procedure**
- Create two LANs.
 - Configure router interfaces.
 - Assign gateways.
 - Ping between PCs.
 - Capture ICMP packets.

IPv4 Header Analysis

Field	Value
Source IP	192.168.1.10
Destination IP	192.168.2.10
TTL	128
Protocol	1 (ICMP)
Header Length	20 Bytes

Calculation
Hop Count = 1 Router
Initial TTL = 128

Final TTL = 127
TTL Reduction = 128 – 127 = **1**

- Result Analysis**
- Router successfully forwards packets.
 - TTL decreases by one after passing through the router.
 - IPv4 header contains routing information.

Conclusion
The Network Layer successfully forwarded packets between different LANs.

Experiment 3: ARP Protocol Analysis

Aim
To study ARP Request and Reply packets.

- Procedure**
1. Create LAN.
 2. Clear ARP cache.
 3. Ping another PC.
 4. Capture ARP packets.

- Observation**
ARP Request
- Sender IP = 192.168.1.10
 - Sender MAC = 00:AA:BB:11:22:33
 - Target IP = 192.168.1.20
 - Target MAC = 00:00:00:00:00:00

- ARP Reply
- Sender IP = 192.168.1.20
 - Sender MAC = 00:AA:BB:44:55:66

Calculation
ARP Requests = 1
ARP Replies = 1
Success Rate
= (Replies / Requests) ×100
= (1/1) ×100
= **100%**

- Result Analysis**
- ARP Request broadcasts the unknown MAC address.
 - ARP Reply returns the destination MAC.
 - Communication starts after ARP resolution.

Conclusion
ARP successfully resolved logical IP addresses into physical MAC addresses.

Experiment 4: ICMP Packet Analysis

Aim
To analyze ICMP Echo Request and Echo Reply messages.

- Procedure**
1. Connect two PCs through router.
 2. Configure IP addresses.
 3. Ping destination.
 4. Capture ICMP packets.

Observation

Field	Value
Type	8 (Echo Request)
Reply Type	0
Code	0
Identifier	0x1234
Sequence Number	1

RTT Calculation
Request Time = 0.015 s
Reply Time = 0.020 s
RTT = Reply – Request
= 0.020 – 0.015
= **0.005 s = 5 ms**
Result Analysis

- ICMP verifies connectivity.
- RTT indicates network delay.
- No packet loss observed.

Conclusion

ICMP effectively assists in network troubleshooting and connectivity verification

Experiment 5: HTTP Client-Server Communication

Aim

To create a client-server network and analyze TCP handshake and HTTP communication.

Procedure

1. Configure Web Server.
2. Enable HTTP service.
3. Connect Client PC.
4. Open browser.
5. Capture packets.

Observation

TCP Three-Way Handshake

1. SYN
2. SYN-ACK
3. ACK

HTTP Communication

- HTTP GET Request
- HTTP 200 OK Response

Calculation

TCP Connection Steps = 3

HTTP Request = 1

HTTP Response = 1

Successful Transactions

$= (1/1) \times 100$

= 100%

Result Analysis

- TCP establishes a reliable connection.
- HTTP transfers webpage successfully.
- Acknowledgements ensure reliable delivery and ordered transmission.

Conclusion

TCP provided reliable communication for HTTP traffic through connection establishment, acknowledgements, and error-free data transfer.